

INCIDENT DETECTION REPORT

Scenario: Multiple Failed Microsoft Entra ID Sign-ins – Possible Brute Force

Field	Details
Platform	Microsoft Sentinel
Data Source	Microsoft Entra ID Sign-in Logs
Severity	Medium
Status	Detection Validated
MITRE ATT&CK	T1110 – Brute Force

1. Executive Summary

This scenario demonstrates the detection of multiple failed Microsoft Entra ID sign-in attempts using Microsoft Sentinel.

A controlled test account was used to generate repeated failed authentication attempts. Microsoft Entra ID Sign-in Logs were ingested into a Log Analytics Workspace and analyzed using KQL.

A custom Scheduled Analytics Rule was configured to detect **five or more failed sign-in attempts against the same user within a 30-minute period**. The rule successfully generated alerts and created a Microsoft Sentinel incident containing the affected user account and source IP address.

2. Lab Environment

The following components were used:

- Microsoft Azure
- Microsoft Entra ID
- Log Analytics Workspace
- Microsoft Sentinel
- Kusto Query Language (KQL)
- Microsoft Entra ID SigninLogs

Detection Flow

Failed Sign-in Attempts -> Microsoft Entra ID -> SigninLogs -> Log Analytics Workspace -> KQL Detection -> Sentinel Analytics Rule -> Alert & Incident

3. Environment Setup & Configuration

The CloudSOC monitoring environment was configured using Microsoft Azure services. A Log Analytics Workspace was created and connected to Microsoft Sentinel to provide centralized security monitoring.

Microsoft Entra ID diagnostic settings were configured to send authentication telemetry to the Log Analytics Workspace. The following logs were used for this scenario:

- Microsoft Entra ID Sign-in Logs
- Microsoft Entra ID Audit Logs

After log ingestion was validated, the SigninLogs table was used to analyze authentication activity using KQL.

A dedicated test account was created to safely simulate authentication failures and validate the detection logic.

Setup Flow

Microsoft Entra ID -> Diagnostic Settings -> Log Analytics Workspace -> Microsoft Sentinel -> KQL Detection & Analytics Rules

Attack Simulation

A dedicated Microsoft Entra ID test account was used to simulate repeated authentication failures.

Multiple incorrect password attempts were performed within a short time period.

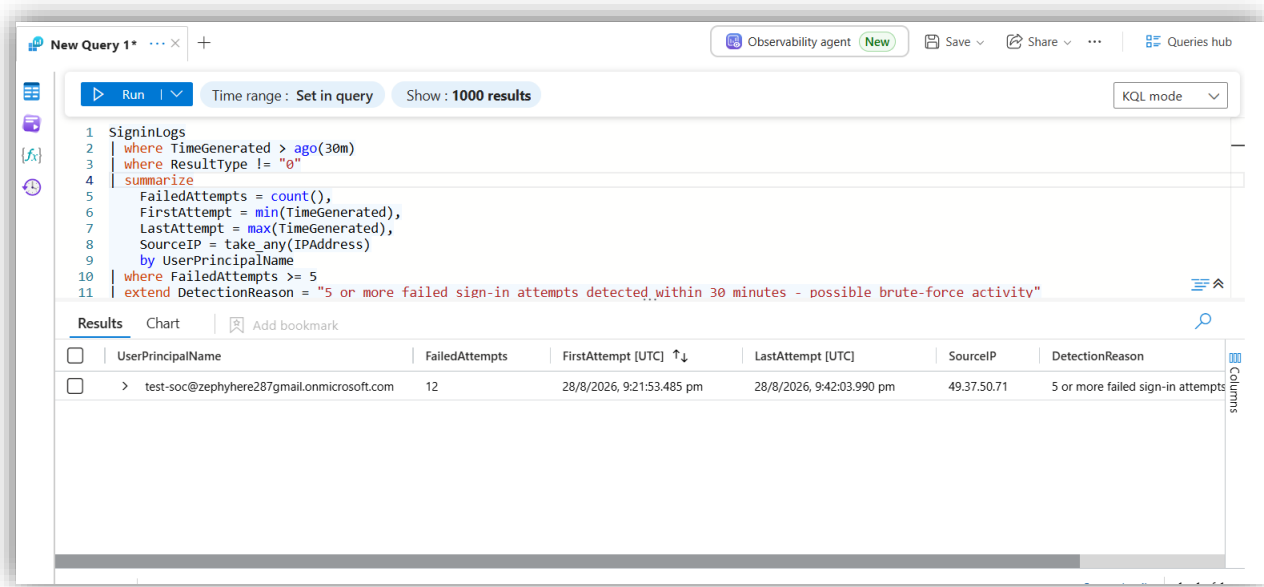
The activity generated authentication events in the SigninLogs table.

During testing, the detection identified:

- **12 failed sign-in attempts**
 - Same user account
 - Activity occurring within the configured detection window
 - Source IP captured for investigation
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4. Detection Logic

The following KQL query was used:



Detection Criteria

Condition	Value
Log Source	SigninLogs
Detection Window	30 Minutes
Activity	Failed Sign-ins
Threshold	5 or more attempts
Grouping	UserPrincipalName

5. Microsoft Sentinel Analytics Rule

A Scheduled Analytics Rule was created with the following configuration:

Setting	Configuration
Rule Name	CloudSOC - Multiple Failed Entra ID Sign-ins

Setting	Configuration
Severity	Medium
Query Frequency	Every 5 Minutes
Lookup Period	30 Minutes
Alert Condition	Query returns results
Incident Creation	Enabled

Entity Mapping

The following entities were mapped:

- **Account:** UserPrincipalName
- **IP Address:** SourceIP

This allowed the affected user and source IP to appear directly within the Sentinel incident.

6. Detection Results

The KQL query successfully identified the test account exceeding the configured threshold.

Microsoft Sentinel subsequently:

1. Executed the scheduled detection query.
2. Identified suspicious authentication activity.
3. Generated security alerts.
4. Created a Sentinel incident.
5. Associated the affected account and source IP as entities.

The incident was generated with **Medium severity**.

During testing, multiple alerts were grouped into the same incident because the detection query ran every five minutes while looking back over a 30-minute period.

7. Investigation Findings

Investigation Item	Finding
Detection	Multiple failed Entra ID sign-ins

Investigation Item	Finding
Failed Attempts	12
Detection Threshold	5
Detection Window	30 minutes
Severity	Medium
Account Entity	Successfully mapped
Source IP	Successfully mapped
Sentinel Incident	Successfully generated

The activity was intentionally generated using a dedicated test account; therefore, no actual account compromise occurred.

The screenshot displays the Microsoft Sentinel interface for an incident titled "CloudSOC - Multiple Failed Entra ID Sign-ins". The incident is categorized as "Medium" severity and "New" status. The workspace is named "law-cloudsoc".

Description: Detects five or more failed Microsoft Entra ID sign-in attempts for the same user within a 30-minute period. This activity may indicate password guessing or brute-force attempts against a cloud identity.

Alert product names: Microsoft Sentinel

Evidence: 2 Events, 2 Alerts, 0 Bookmarks

Last update time: 8/29/2026, 3:09:49 AM
Creation time: 8/29/2026, 3:08:28 AM

Incident timeline: Shows two entries for "CloudSOC - Multipl..." on 29 Aug 02:33:40 and 29 Aug 02:32:45.

Entities: Includes "test-soc@zephyr Account" and "49.37 IP".

Top insights: Shows "Actions by account" for "test-soc@zephyr" on 8/28/2026, 3:08:28 AM - 8/29/2026, 3:09:32 AM.

Similar incidents: A table showing one incident with severity "Medium", incident number "1", title "CloudSOC - Multiple Failed Sign...", and last update time "29/8/2026, 03:00 am".

8. MITRE ATT&CK Mapping

Technique

T1110 – Brute Force

Repeated authentication failures may indicate password guessing or brute-force activity targeting a cloud identity.

9. Recommendations

The detection can be improved further by:

- Filtering specific Entra ID authentication failure codes.
 - Detecting failed sign-ins followed by a successful authentication.
 - Adding IP reputation checks.
 - Comparing sign-ins against normal user behavior.
 - Tuning alert suppression to reduce duplicate alerts.
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10. Conclusion

This scenario successfully demonstrated an end-to-end cloud identity detection workflow using Microsoft Sentinel.

Repeated failed Microsoft Entra ID sign-in attempts were simulated, collected through SigninLogs, detected using KQL, and converted into a Sentinel Analytics Rule.

The detection successfully generated alerts and created a security incident with relevant entities for investigation.

This scenario represents the first completed detection use case in the **CloudSOC Lab project**.